AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology
Department of Computer Science
CSC 3215: Web Technologies

Semester: Summer 2025-26 **Group No: 01** **Section:** N

Group Members:

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2	23-53907-3	MUNTASIRUL HOQUE	Member	
3	23-53388-3	FARAH IBNAT JHUMU	Member	
4	23-55613-3	TASNIA ISLAM	Member	

Project Title

WorkNest -- Real-Time Workforce Analytics & Management System

Short Description

WorkNest is a web-based Employee Management System designed to centralize employee information, manage workforce activities, and streamline HR and managerial workflows through a user-friendly interface.

Introduction

In a software development company, managing employees, work hours, tasks, leave, and promotions can be challenging without an organized digital platform. WorkNest provides a centralized web-based system where HR personnel, managers, and employees can securely manage and access relevant workforce information. The system reduces manual work, improves data organization, and supports efficient decision-making.

Background Study

Traditional employee management processes often depend on manual records and disconnected workflows, which can cause data errors, delays, and difficulty in tracking employee activities. The existing Employee Management System project addresses these issues using a centralized database, role-based access control, and workflow management for recruitment, termination, promotion, leave, and task assignment. WorkNest adapts these core requirements into a web-based application suitable for the Web Technologies course.

GitHub: GitHub - rafidhossain404/EmployeeManagementSystem_WebtechSummer2025-2026

ER Diagram

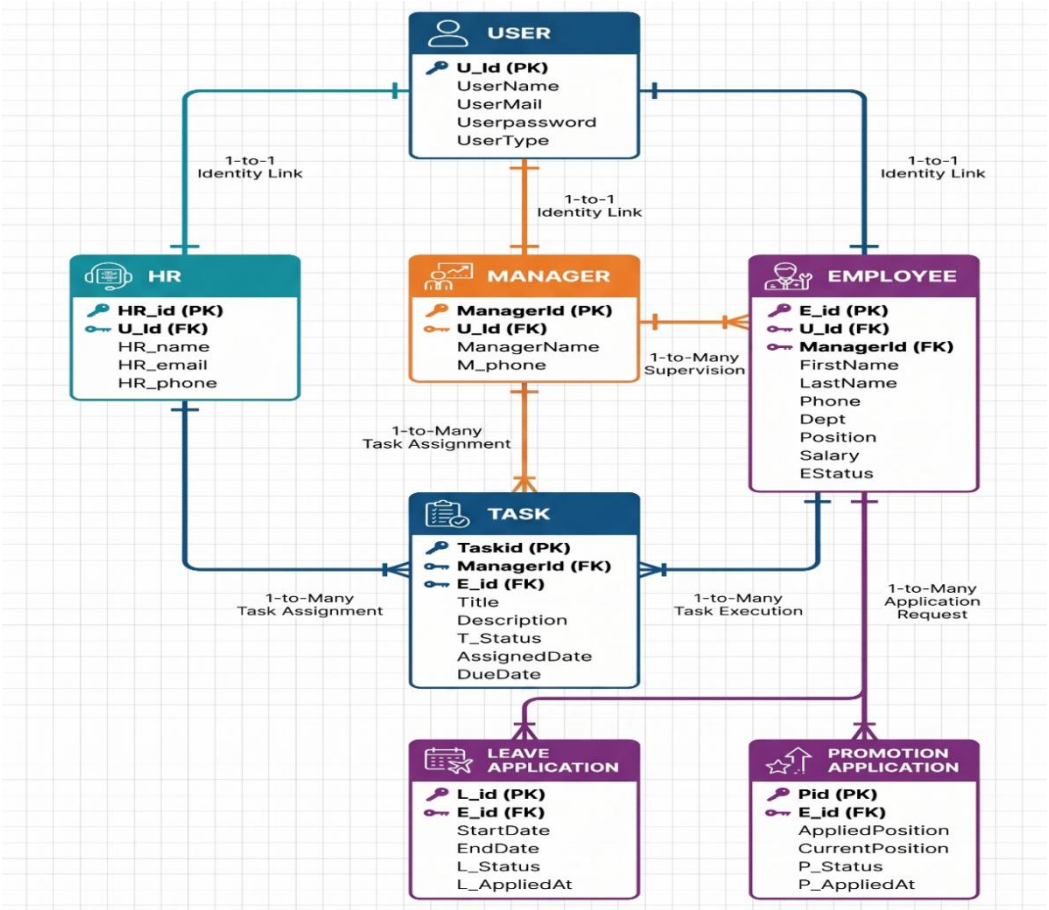


Figure 1 — Entity-relationship diagram of the WorkNest -- Real-Time Workforce Analytics & Management System

Use Case Diagram

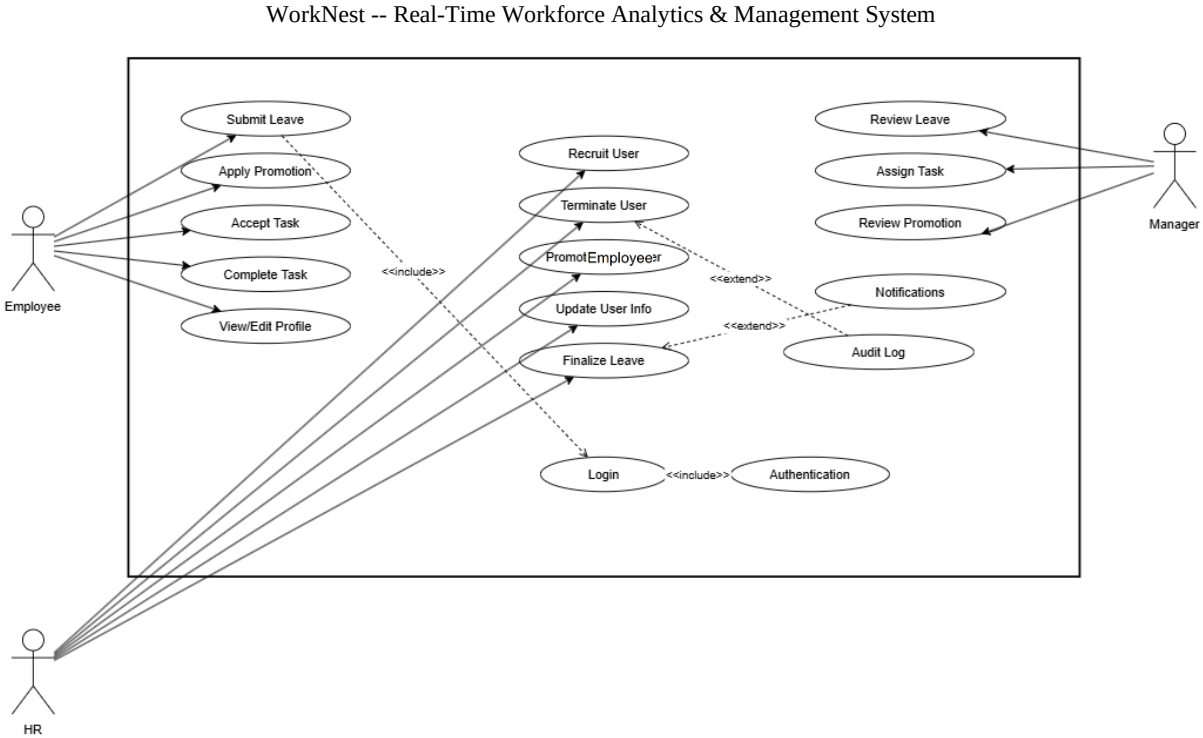
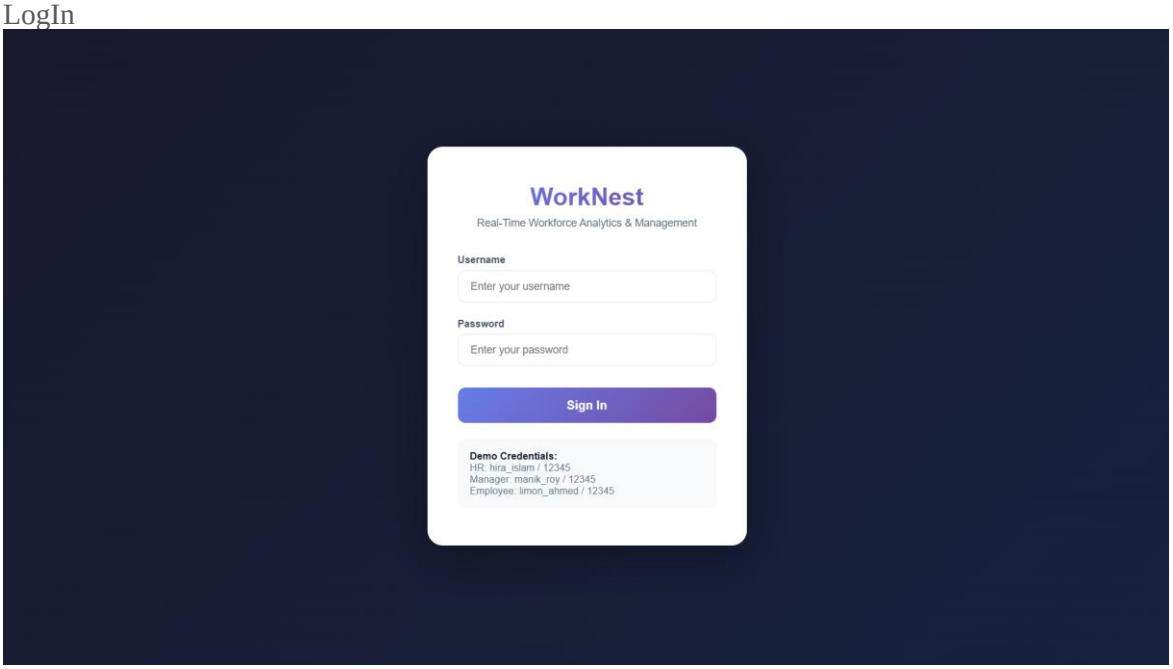
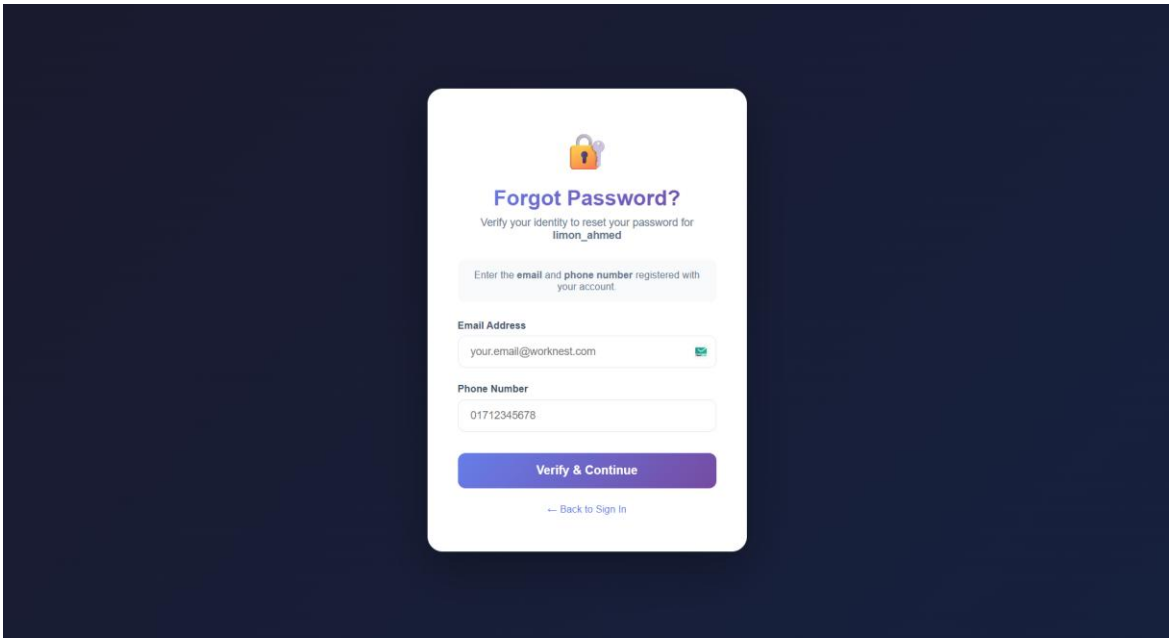


Figure 2 — Use case diagram of the WorkNest -- Real-Time Workforce Analytics & Management System

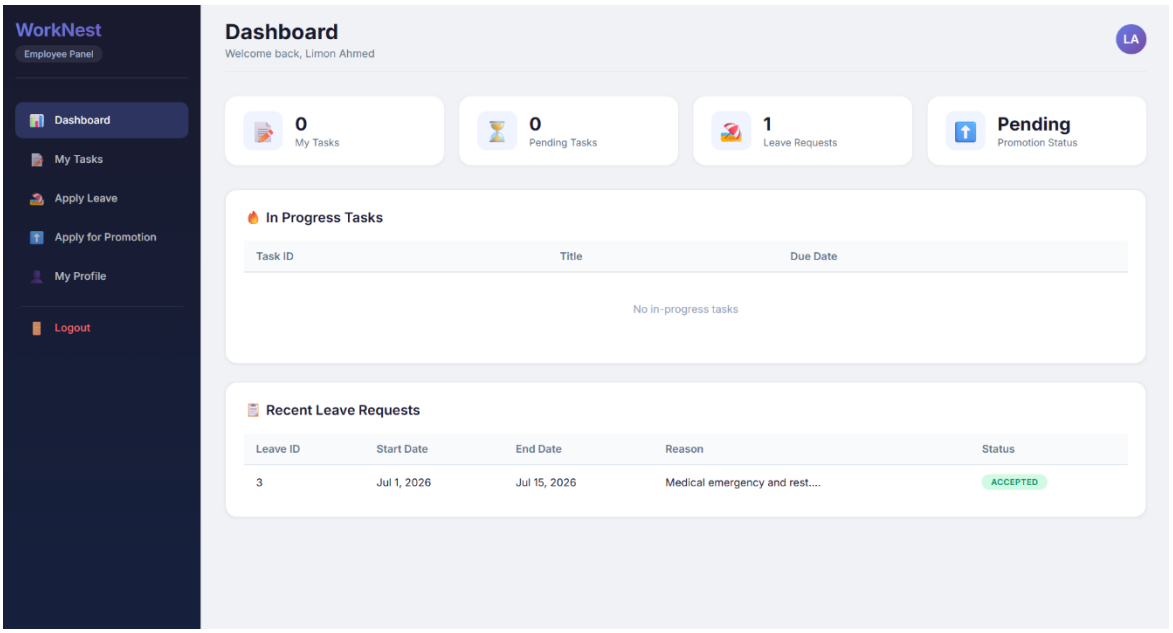
UI (Figma/Sketch)

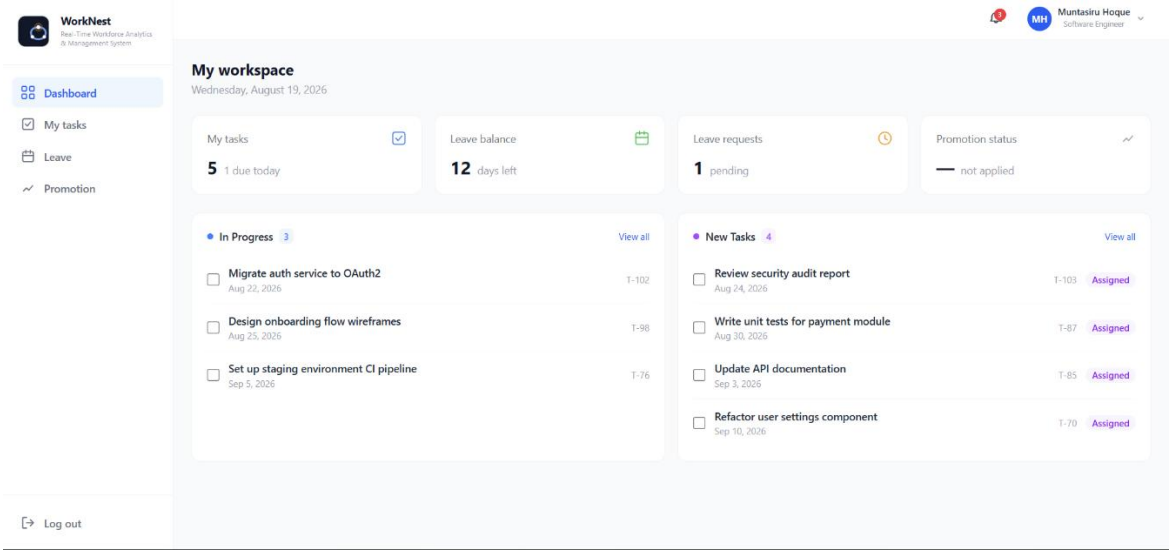
LogIn

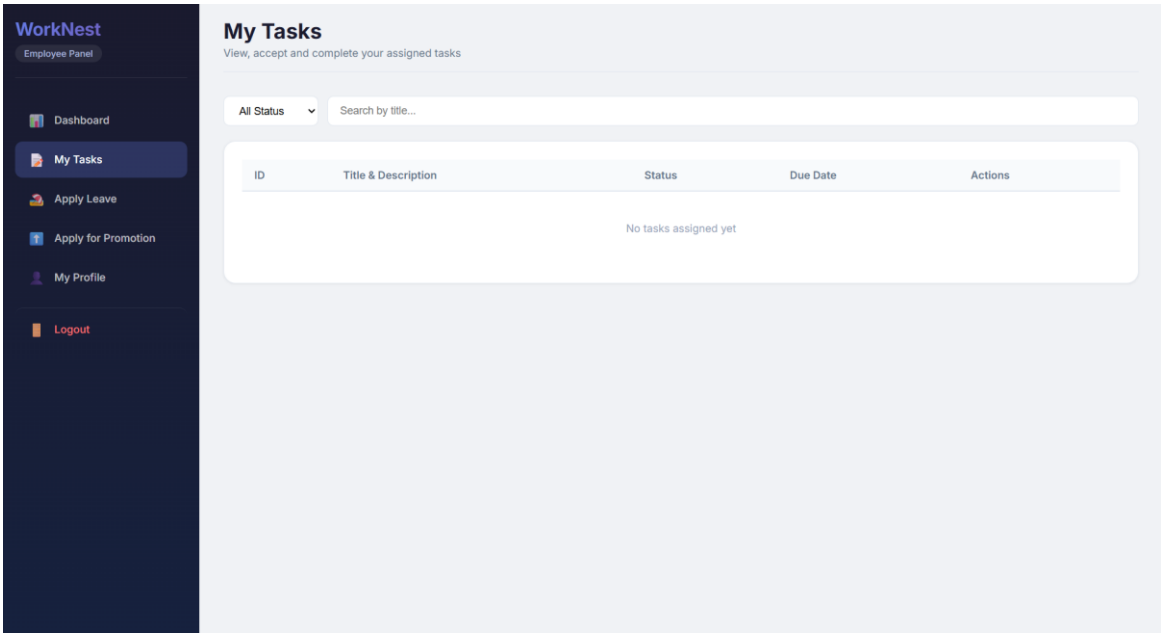


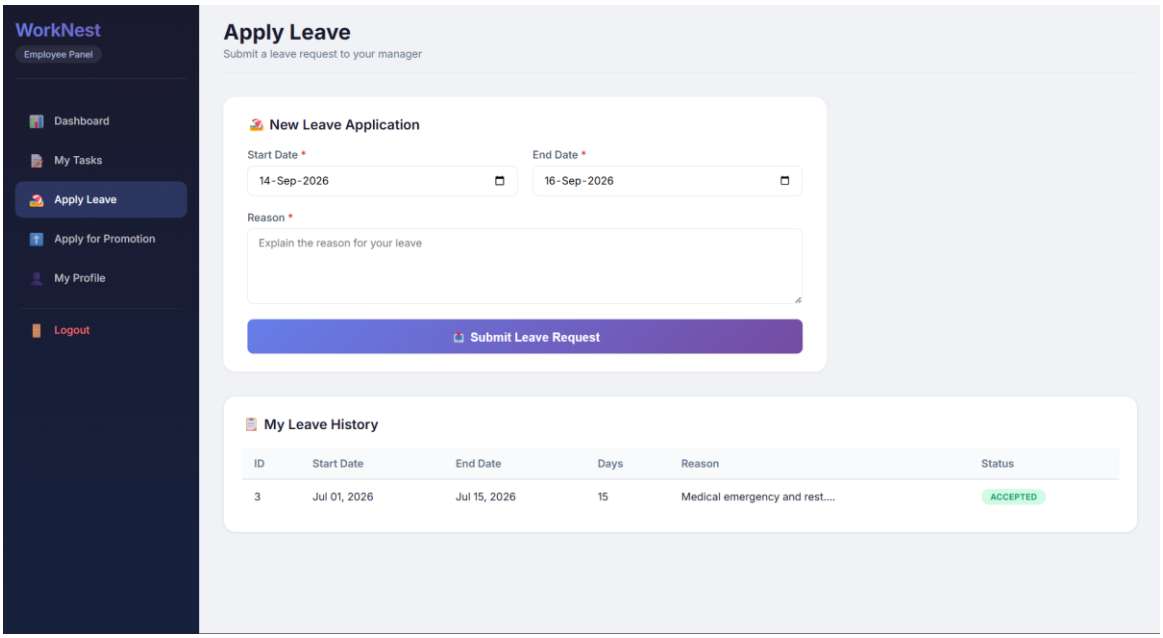


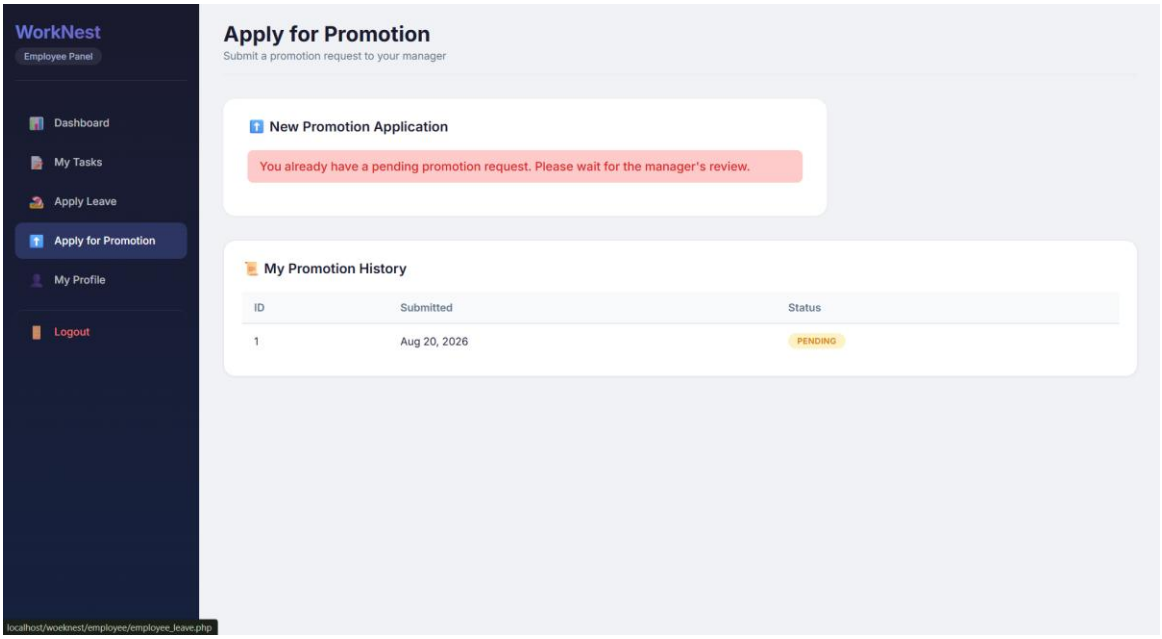
Employee

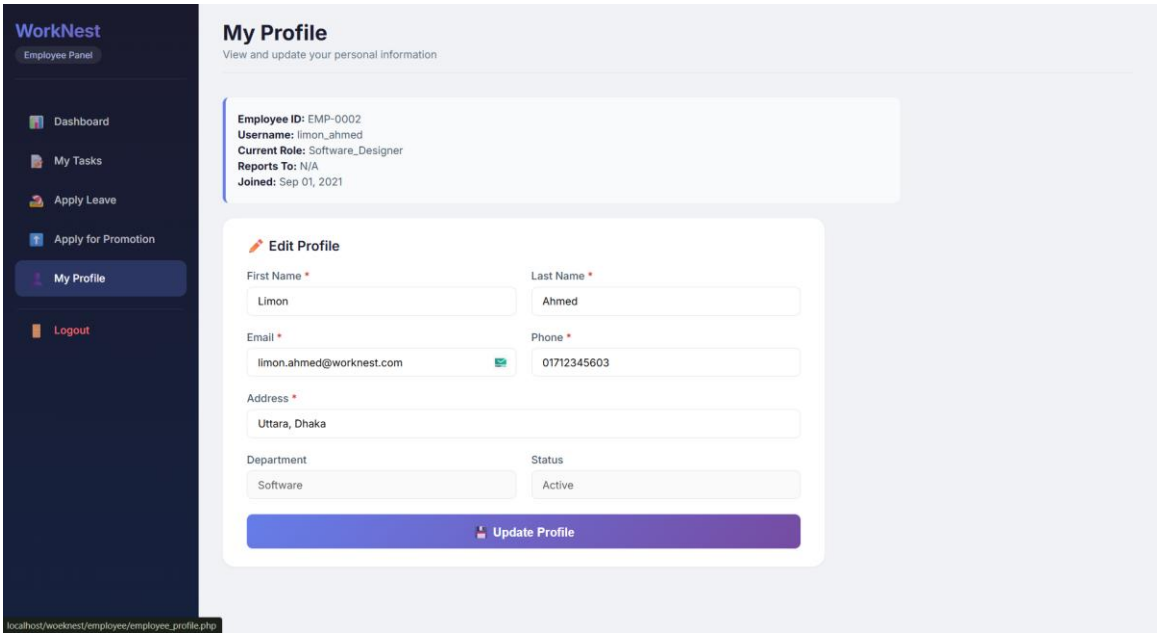



WorkNest
Real-Time Workforce Analytics & Management System
Muntasiru Hoque
Software Engineer
Dashboard
My tasks
Leave
Promotion
Log out
My workspace
Wednesday, August 19, 2026
My tasks
5 1 due today
Leave balance
12 days left
Leave requests
1 pending
Promotion status
— not applied
In Progress 3
View all
Migrate auth service to OAuth2
Aug 22, 2026
T-102
Design onboarding flow wireframes
Aug 25, 2026
T-98
Set up staging environment CI pipeline
Sep 5, 2026
T-76
New Tasks 4
View all
Review security audit report
Aug 24, 2026
T-103
Assigned
Write unit tests for payment module
Aug 30, 2026
T-87
Assigned
Update API documentation
Sep 3, 2026
T-85
Assigned
Refactor user settings component
Sep 10, 2026
T-70
Assigned

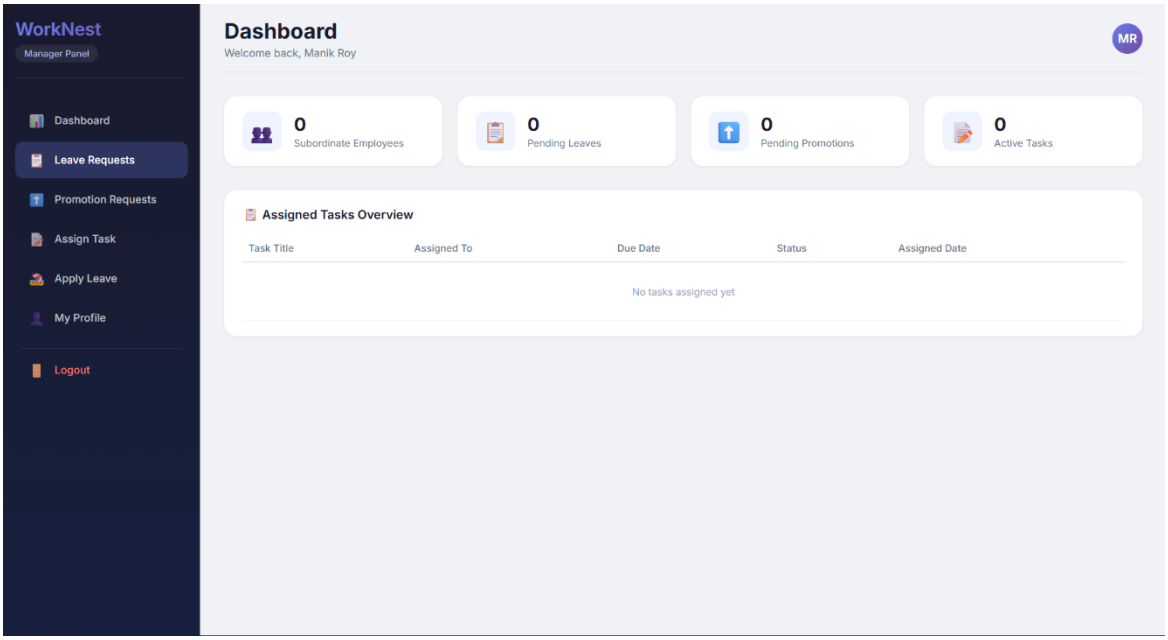

WorkNest
Employee Panel
Dashboard
My Tasks
Apply Leave
Apply for Promotion
My Profile
Logout
My Tasks
View, accept and complete your assigned tasks
All Status
Search by title...
ID
Title & Description
Status
Due Date
Actions
No tasks assigned yet

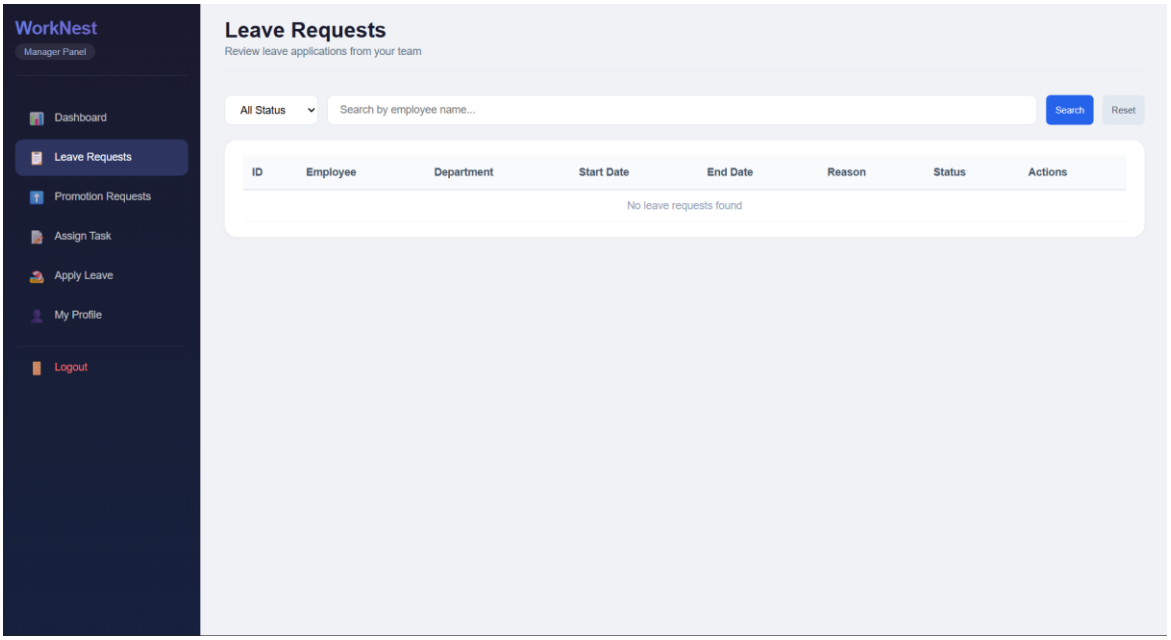

WorkNest
Employee Panel
Dashboard
My Tasks
Apply Leave
Apply for Promotion
My Profile
Logout
Apply Leave
Submit a leave request to your manager
New Leave Application
Start Date *
14-Sep-2026
End Date *
16-Sep-2026
Reason *
Explain the reason for your leave
Submit Leave Request
My Leave History
ID
Start Date
End Date
Days
Reason
Status
3
Jul 01, 2026
Jul 15, 2026
15
Medical emergency and rest....
ACCEPTED

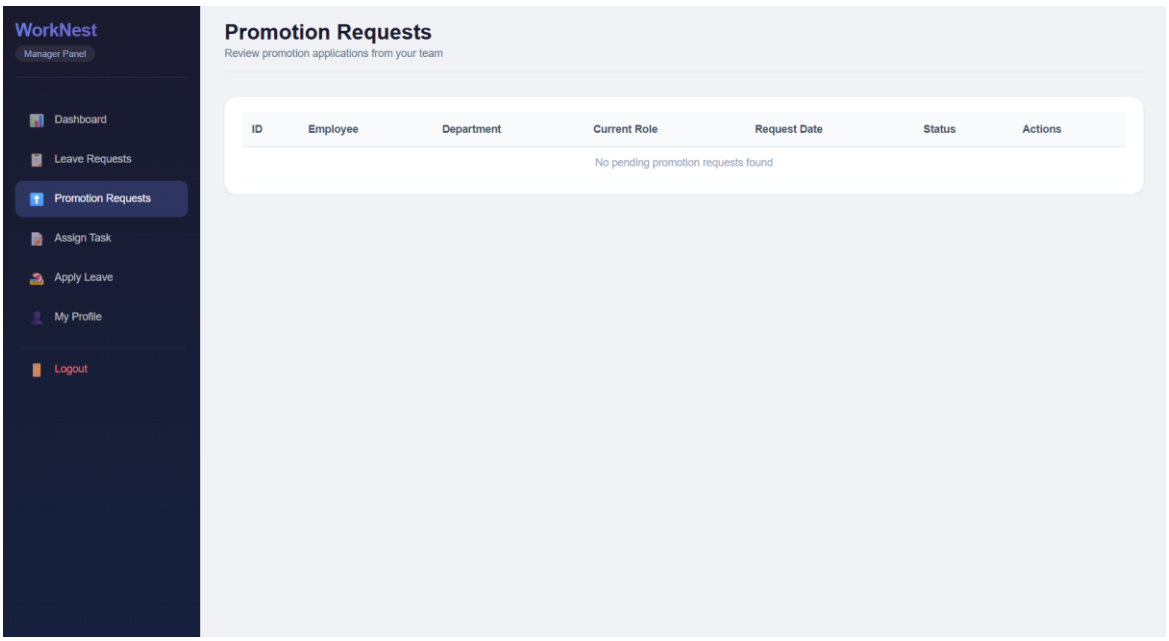

WorkNest
Employee Panel
Dashboard
My Tasks
Apply Leave
Apply for Promotion
My Profile
Logout
Apply for Promotion
Submit a promotion request to your manager
New Promotion Application
You already have a pending promotion request. Please wait for the manager's review.
My Promotion History
ID
Submitted
Status
1
Aug 20, 2026
PENDING
localhost/woeknest/employee/employee_leave.php

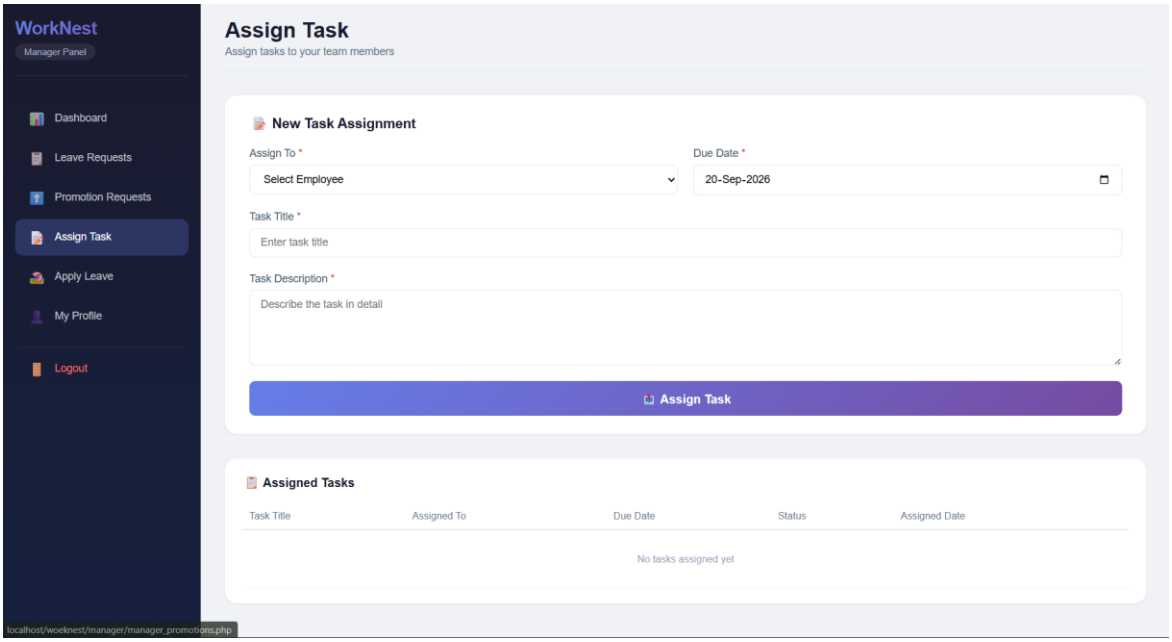

WorkNest
Employee Panel
Dashboard
My Tasks
Apply Leave
Apply for Promotion
My Profile
Logout
My Profile
View and update your personal information
Employee ID: EMP-0002
Username: limon_ahmed
Current Role: Software_Designer
Reports To: N/A
Joined: Sep 01, 2021
Edit Profile
First Name *
Limon
Last Name *
Ahmed
Email *
limon.ahmed@worknest.com
Phone *
01712345603
Address *
Uttara, Dhaka
Department
Software
Status
Active
Update Profile
localhost/woeknest/employee/employee_profile.php

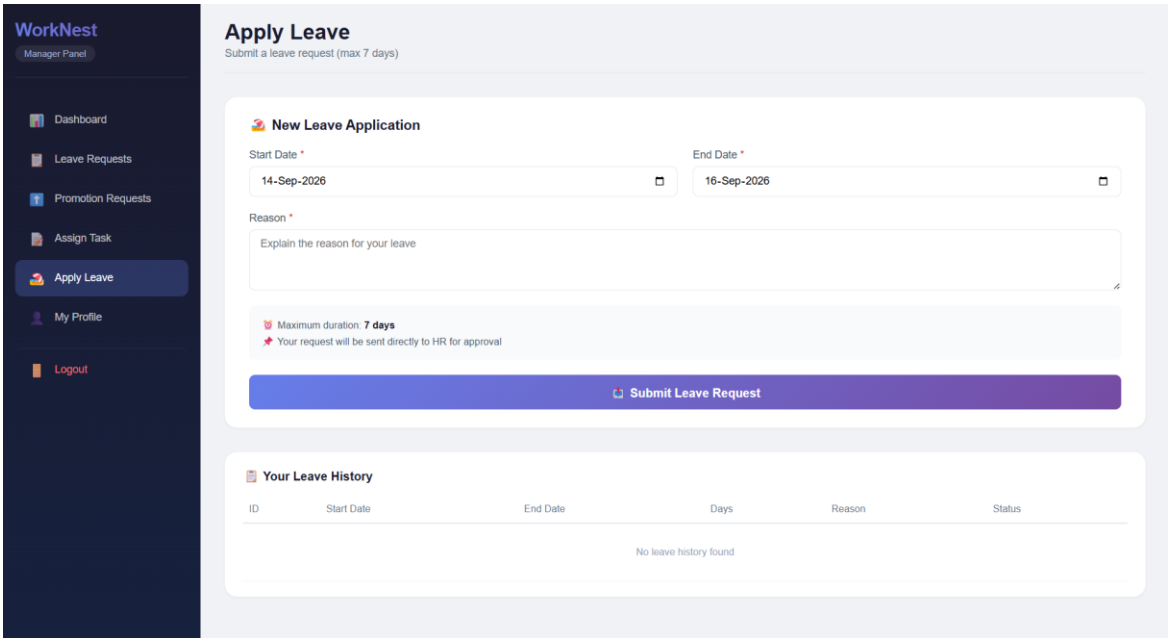
Manager

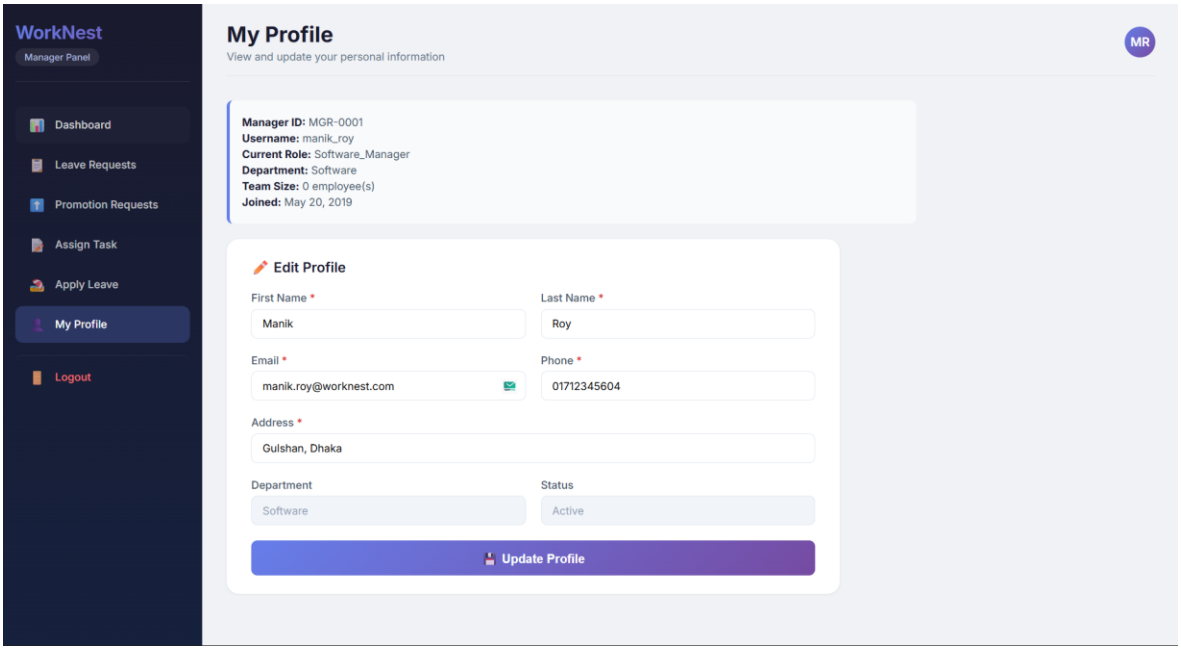

WorkNest
Manager Panel
Dashboard
Leave Requests
Promotion Requests
Assign Task
Apply Leave
My Profile
Logout
Dashboard
Welcome back, Manik Roy
MR
0
Subordinate Employees
0
Pending Leaves
0
Pending Promotions
0
Active Tasks
Assigned Tasks Overview
Task Title
Assigned To
Due Date
Status
Assigned Date
No tasks assigned yet


WorkNest
Manager Panel
Dashboard
Leave Requests
Promotion Requests
Assign Task
Apply Leave
My Profile
Logout
Leave Requests
Review leave applications from your team
All Status
Search by employee name...
Search
Reset
ID
Employee
Department
Start Date
End Date
Reason
Status
Actions
No leave requests found

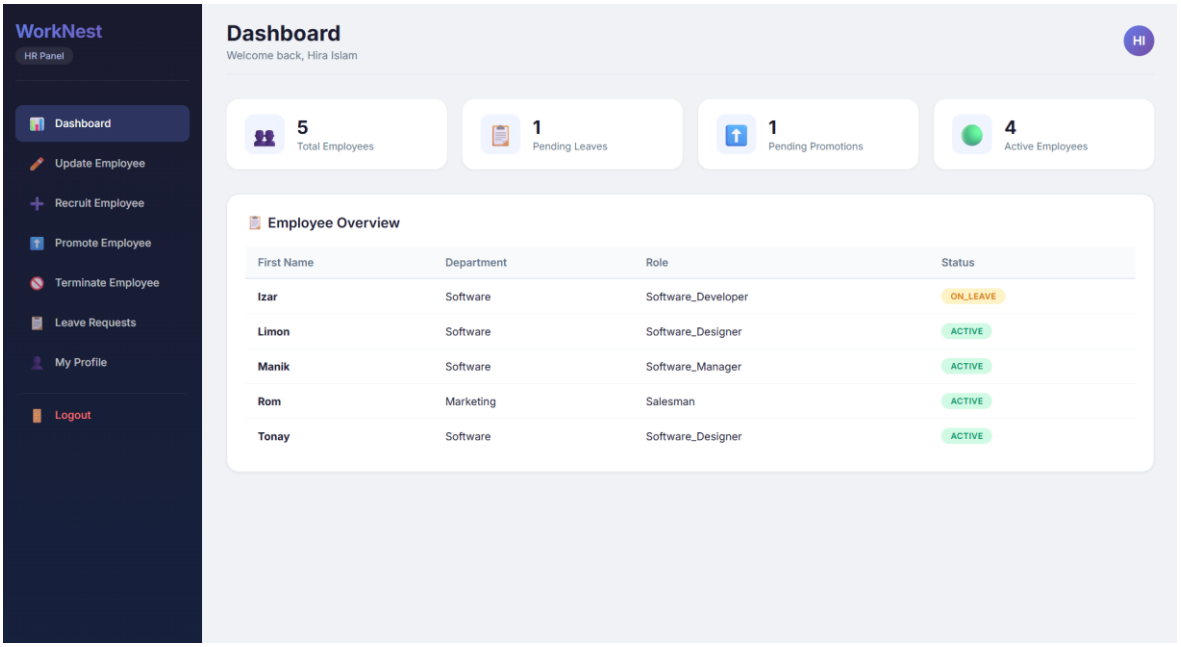

WorkNest
Manager Panel
Dashboard
Leave Requests
Promotion Requests
Assign Task
Apply Leave
My Profile
Logout
Promotion Requests
Review promotion applications from your team
ID
Employee
Department
Current Role
Request Date
Status
Actions
No pending promotion requests found

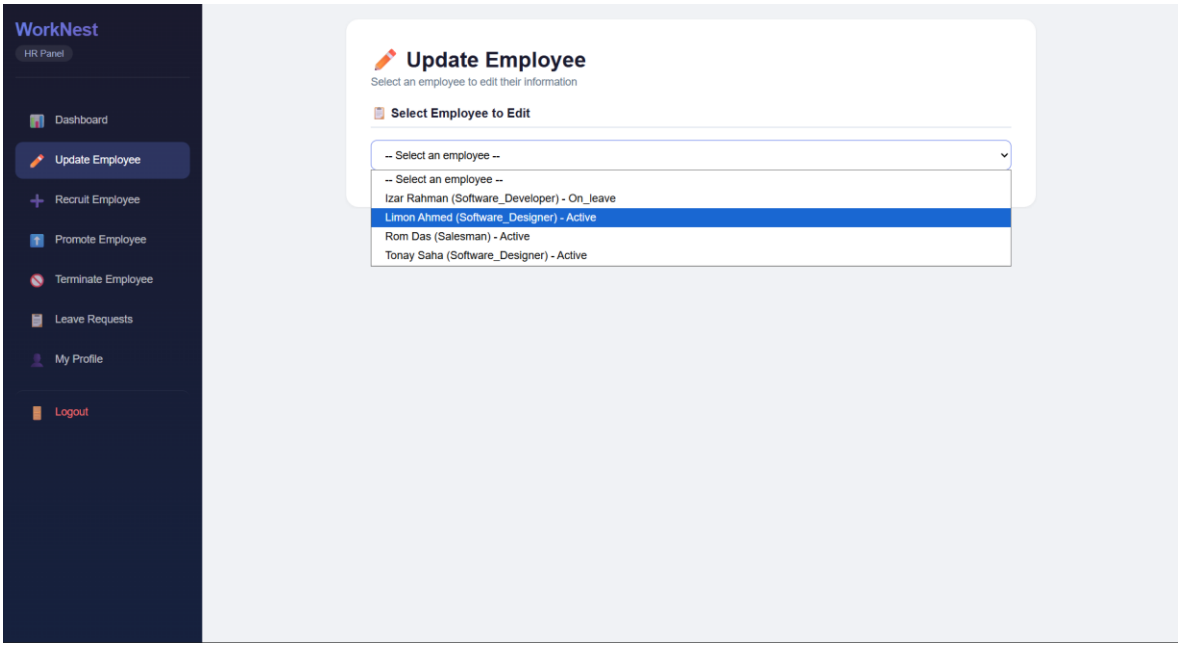

WorkNest
Manager Panel
Dashboard
Leave Requests
Promotion Requests
Assign Task
Apply Leave
My Profile
Logout
Assign Task
Assign tasks to your team members
New Task Assignment
Assign To *
Select Employee
Due Date *
20-Sep-2026
Task Title *
Enter task title
Task Description *
Describe the task in detail
Assign Task
Assigned Tasks
Task Title
Assigned To
Due Date
Status
Assigned Date
No tasks assigned yet
localhost/woeknest/manager/manager_promotions.php

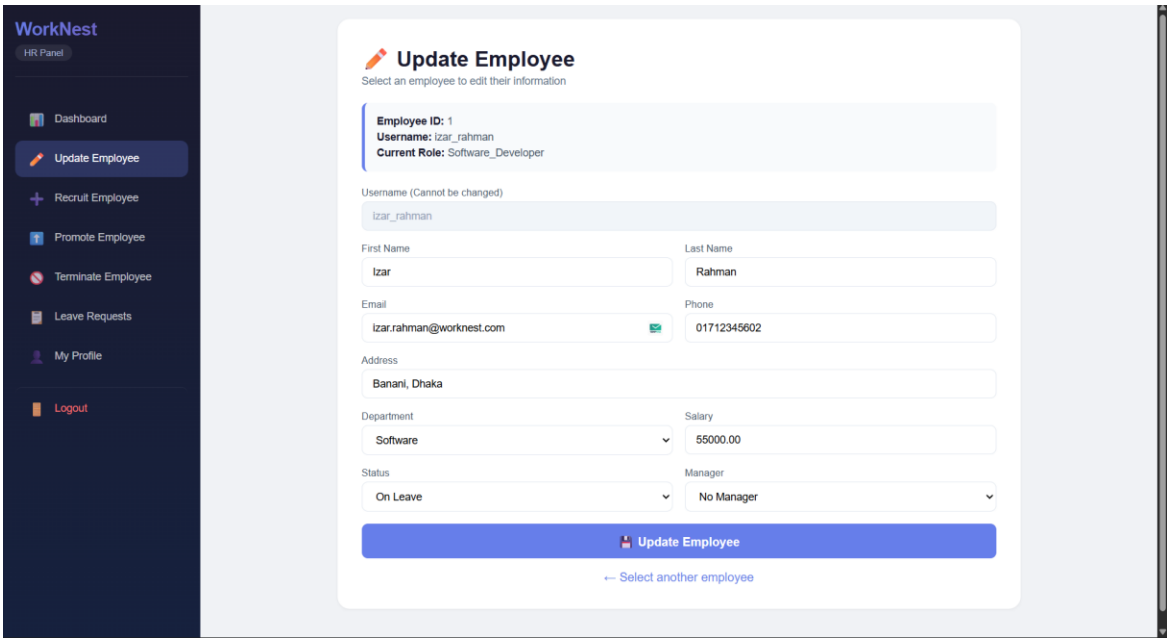

WorkNest
Manager Panel
Dashboard
Leave Requests
Promotion Requests
Assign Task
Apply Leave
My Profile
Logout
Apply Leave
Submit a leave request (max 7 days)
New Leave Application
Start Date *
14-Sep-2026
End Date *
16-Sep-2026
Reason *
Explain the reason for your leave
Maximum duration: 7 days
Your request will be sent directly to HR for approval
Submit Leave Request
Your Leave History
ID
Start Date
End Date
Days
Reason
Status
No leave history found

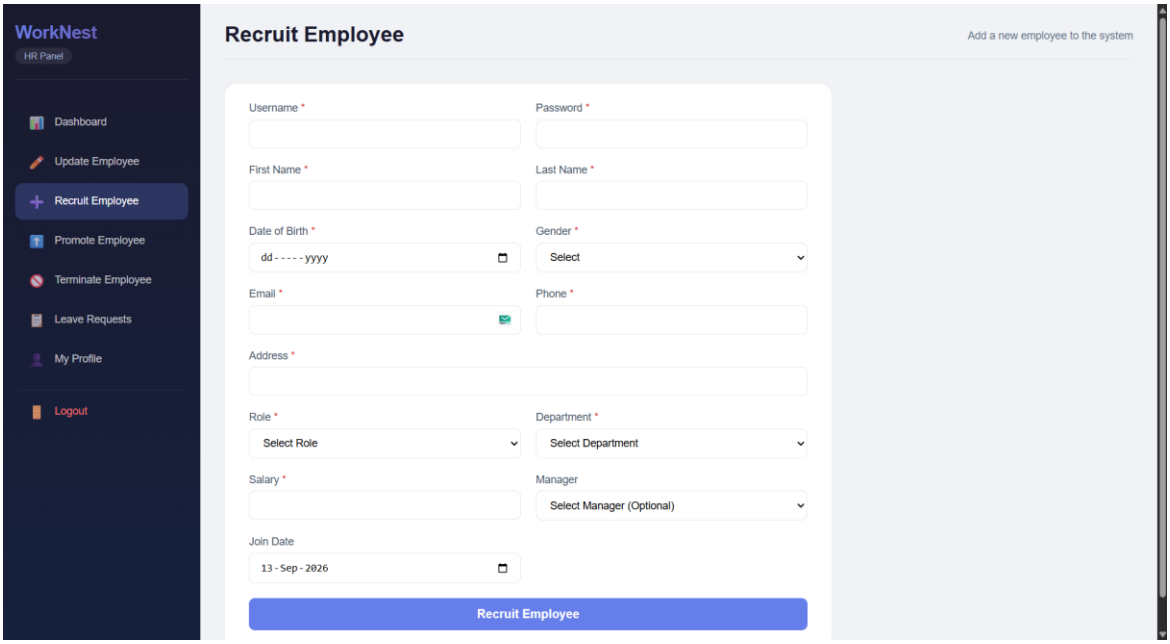

WorkNest
Manager Panel
Dashboard
Leave Requests
Promotion Requests
Assign Task
Apply Leave
My Profile
Logout
My Profile
View and update your personal information
MR
Manager ID: MGR-0001
Username: manik_roy
Current Role: Software_Manager
Department: Software
Team Size: 0 employee(s)
Joined: May 20, 2019
Edit Profile
First Name *
Manik
Last Name *
Roy
Email *
manik.roy@worknest.com
Phone *
01712345604
Address *
Gulshan, Dhaka
Department
Software
Status
Active
Update Profile

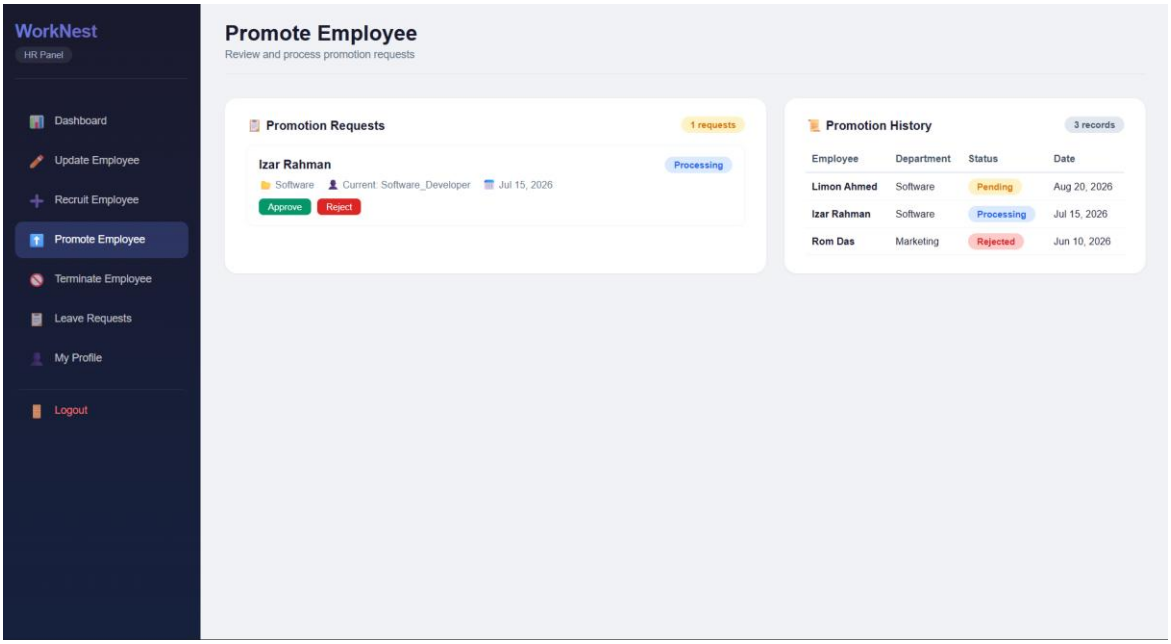
HR

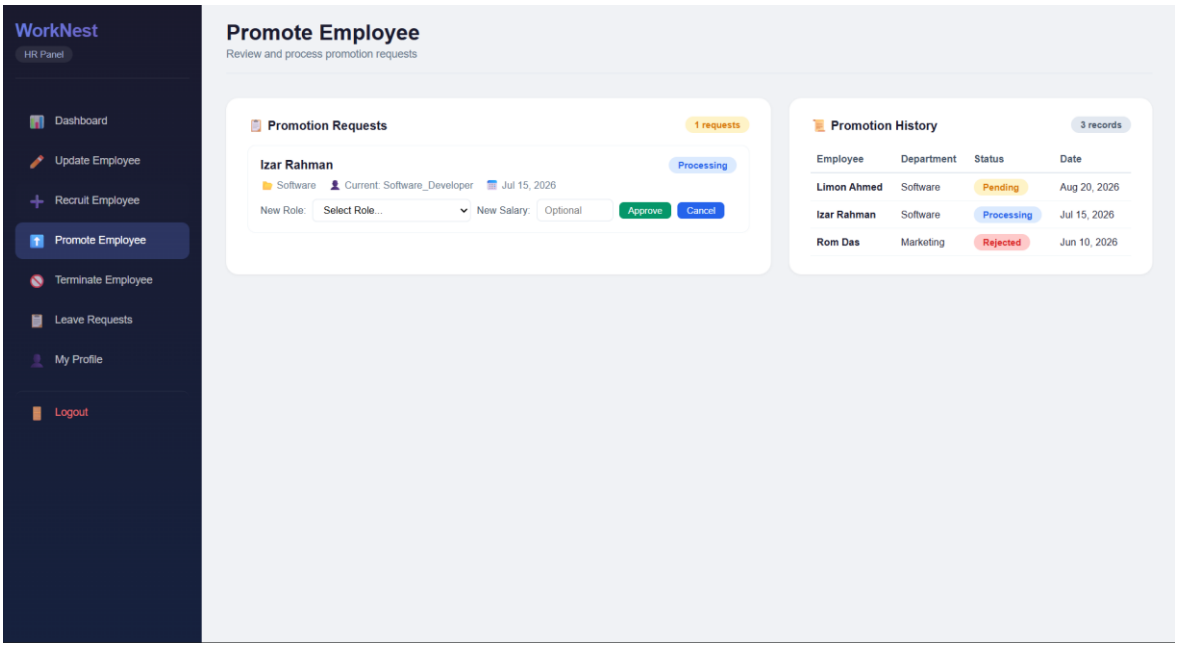

WorkNest
HR Panel
Dashboard
Update Employee
Recruit Employee
Promote Employee
Terminate Employee
Leave Requests
My Profile
Logout
Dashboard
Welcome back, Hira Islam
HI
5
Total Employees
1
Pending Leaves
1
Pending Promotions
4
Active Employees
Employee Overview
First Name
Department
Role
Status
Izar
Software
Software_Developer
ON_LEAVE
Limon
Software
Software_Designer
ACTIVE
Manik
Software
Software_Manager
ACTIVE
Rom
Marketing
Salesman
ACTIVE
Tonay
Software
Software_Designer
ACTIVE

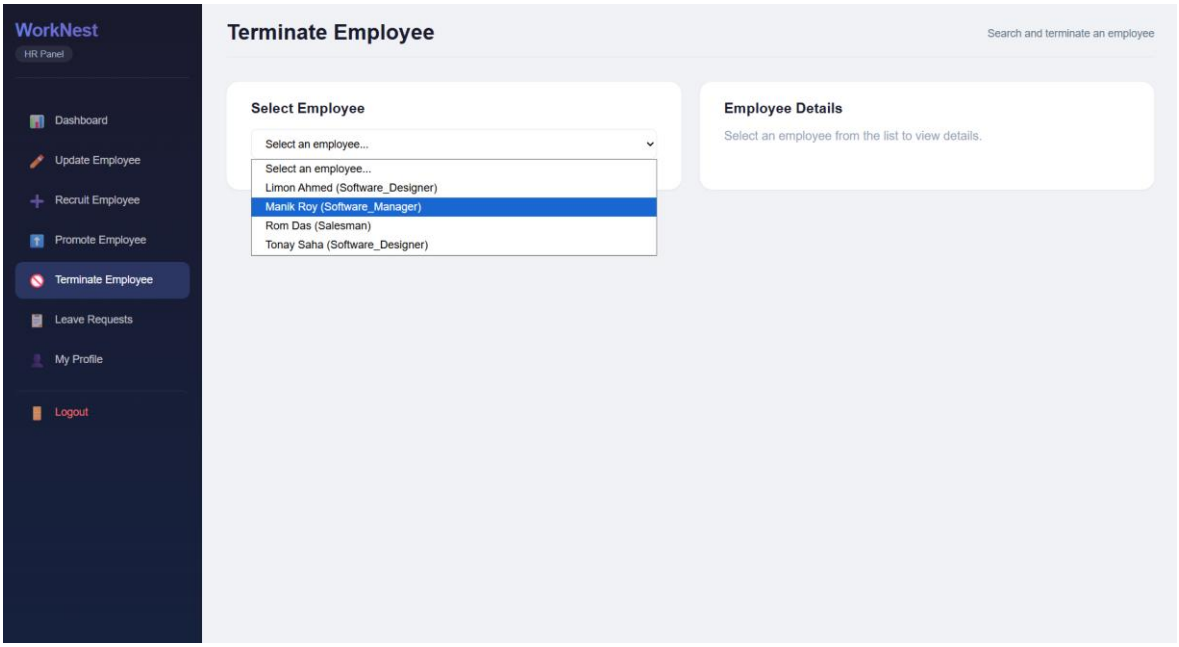

WorkNest
HR Panel
Dashboard
Update Employee
Recruit Employee
Promote Employee
Terminate Employee
Leave Requests
My Profile
Logout
Update Employee
Select an employee to edit their information
Select Employee to Edit
-- Select an employee --
-- Select an employee --
Izar Rahman (Software_Developer) - On_leave
Limon Ahmed (Software_Designer) - Active
Rom Das (Salesman) - Active
Tonay Saha (Software_Designer) - Active

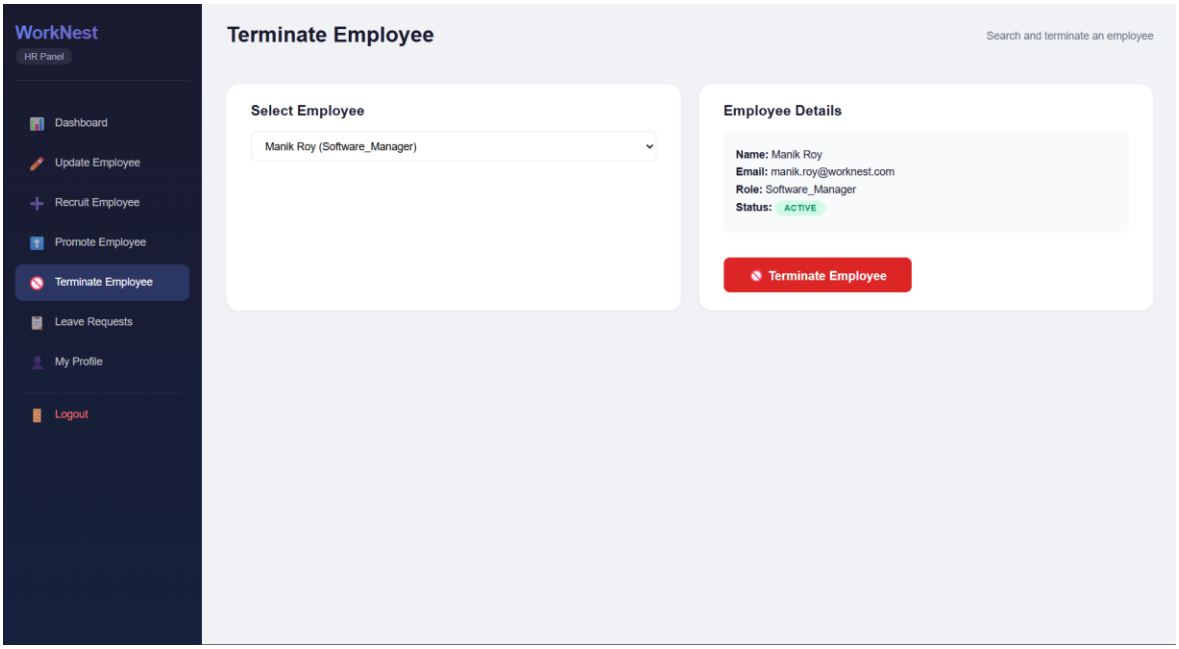

WorkNest
HR Panel
Dashboard
Update Employee
Recruit Employee
Promote Employee
Terminate Employee
Leave Requests
My Profile
Logout
Update Employee
Select an employee to edit their information
Employee ID: 1
Username: izar_rahman
Current Role: Software_Developer
Username (Cannot be changed)
izar_rahman
First Name
Izar
Last Name
Rahman
Email
izar.rahman@worknest.com
Phone
01712345602
Address
Banani, Dhaka
Department
Software
Salary
55000.00
Status
On Leave
Manager
No Manager
Update Employee
← Select another employee

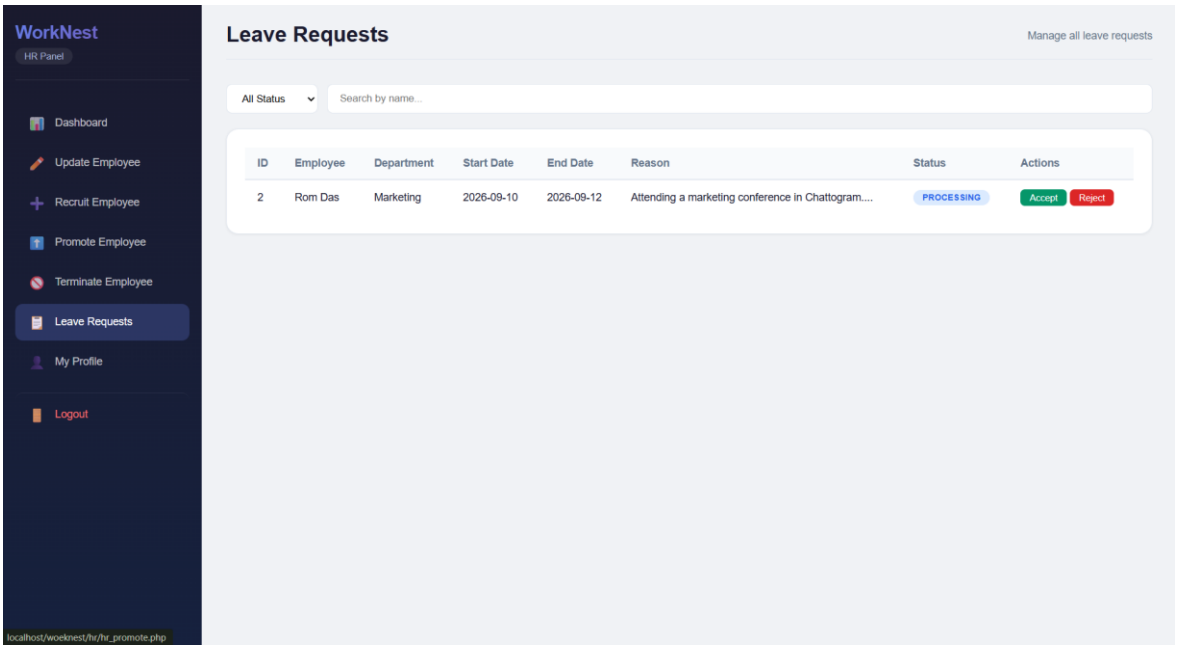

WorkNest
HR Panel
Dashboard
Update Employee
Recruit Employee
Promote Employee
Terminate Employee
Leave Requests
My Profile
Logout
Recruit Employee
Add a new employee to the system
Username *
Password *
First Name *
Last Name *
Date of Birth *
dd - - - - yyyy
Gender *
Select
Email *
Phone *
Address *
Role *
Select Role
Department *
Select Department
Salary *
Manager
Select Manager (Optional)
Join Date
13 - Sep - 2026
Recruit Employee

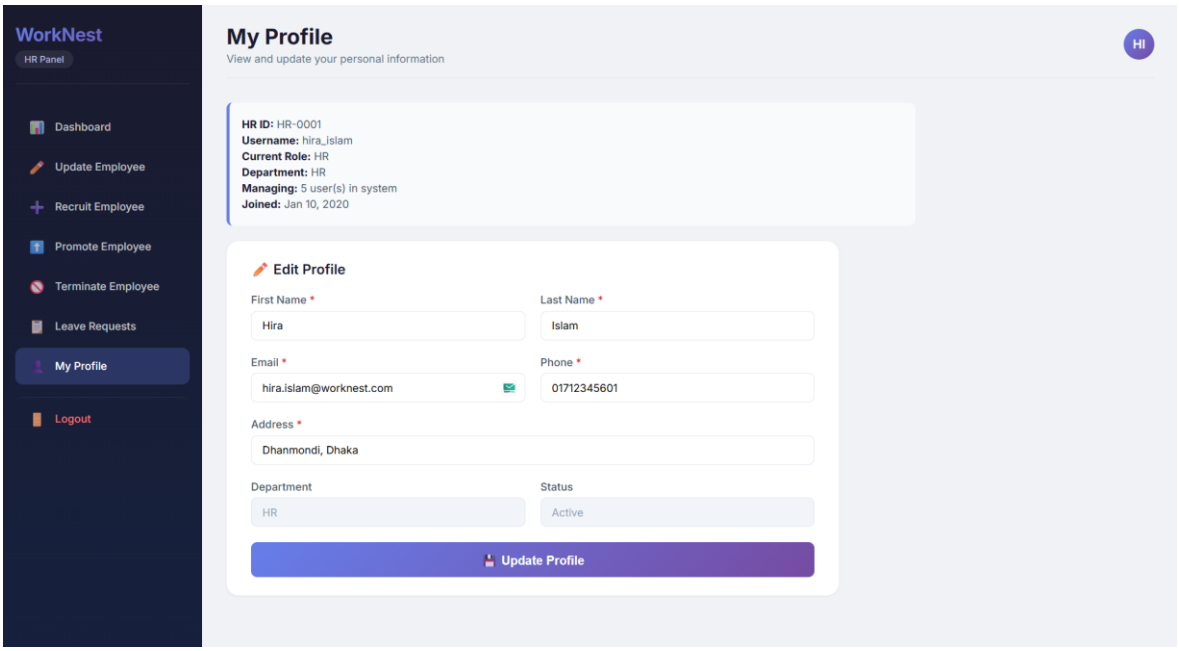

WorkNest
HR Panel
Dashboard
Update Employee
Recruit Employee
Promote Employee
Terminate Employee
Leave Requests
My Profile
Logout
Promote Employee
Review and process promotion requests
Promotion Requests
1 requests
Izar Rahman
Processing
Software
Current: Software_Developer
Jul 15, 2026
Approve
Reject
Promotion History
3 records
Employee
Department
Status
Date
Limon Ahmed
Software
Pending
Aug 20, 2026
Izar Rahman
Software
Processing
Jul 15, 2026
Rom Das
Marketing
Rejected
Jun 10, 2026


WorkNest
HR Panel
Dashboard
Update Employee
Recruit Employee
Promote Employee
Terminate Employee
Leave Requests
My Profile
Logout
Promote Employee
Review and process promotion requests
Promotion Requests
1 requests
Izar Rahman
Processing
Software
Current: Software_Developer
Jul 15, 2026
New Role:
Select Role...
New Salary:
Optional
Approve
Cancel
Promotion History
3 records
Employee
Department
Status
Date
Limon Ahmed
Software
Pending
Aug 20, 2026
Izar Rahman
Software
Processing
Jul 15, 2026
Rom Das
Marketing
Rejected
Jun 10, 2026


WorkNest
HR Panel
Dashboard
Update Employee
Recruit Employee
Promote Employee
Terminate Employee
Leave Requests
My Profile
Logout
Terminate Employee
Search and terminate an employee
Select Employee
Select an employee...
Select an employee...
Limon Ahmed (Software_Designer)
Manik Roy (Software_Manager)
Rom Das (Salesman)
Tonay Saha (Software_Designer)
Employee Details
Select an employee from the list to view details.


WorkNest
HR Panel
Dashboard
Update Employee
Recruit Employee
Promote Employee
Terminate Employee
Leave Requests
My Profile
Logout
Terminate Employee
Search and terminate an employee
Select Employee
Manik Roy (Software_Manager)
Employee Details
Name: Manik Roy
Email: manik.roy@worknest.com
Role: Software_Manager
Status: ACTIVE
Terminate Employee


WorkNest
HR Panel
Dashboard
Update Employee
Recruit Employee
Promote Employee
Terminate Employee
Leave Requests
My Profile
Logout
Leave Requests
Manage all leave requests
All Status
Search by name...
ID
Employee
Department
Start Date
End Date
Reason
Status
Actions
2
Rom Das
Marketing
2026-09-10
2026-09-12
Attending a marketing conference in Chattogram....
PROCESSING
Accept
Reject
localhost/woeknest/hr/hr_promote.php


WorkNest
HR Panel
Dashboard
Update Employee
Recruit Employee
Promote Employee
Terminate Employee
Leave Requests
My Profile
Logout
My Profile
View and update your personal information
HI
HR ID: HR-0001
Username: hira_islam
Current Role: HR
Department: HR
Managing: 5 user(s) in system
Joined: Jan 10, 2020
Edit Profile
First Name *
Hira
Last Name *
Islam
Email *
hira.islam@worknest.com
Phone *
01712345601
Address *
Dhanmondi, Dhaka
Department
HR
Status
Active
Update Profile

Types of Users: 1. HR 2. Manager 3. Employee

Common Features (Available to All Users)

- **Authentication**
  1. Login to the system
  2. Logout from the system
  3. User registration
- **Account Management**
  1. Change or reset password
  2. Manage profile information (view, edit, delete)
- **Dashboard**
  1. Access a personalized dashboard after login

Specific Features by User Type

- **Features of Employee**
  1. **Apply for Leave** -- Submit leave requests to their manager.
  2. **Apply for Promotion** -- Request promotion to Manager through the manager → HR workflow.
  3. **Complete Assigned Tasks** -- Accept tasks and update their status to *In Progress* and then *Complete*.
  4. **View Assigned Tasks** -- Access assigned task details, instructions, and deadlines.

- **Features of Manager**
  1. **Review Leave Requests** -- Review leaves applications submitted by employees.
  2. **Review Promotion Requests** -- Review employee promotion applications and forward approved requests to HR.
  3. **Apply for Leave** -- Submit a leave request directly to HR
  4. **Assign Tasks** -- Assign tasks to employees with instructions and deadlines.
  5. **Track Task Progress** -- Monitor tasks assigned to employees.

- **Features of HR**
  1. **Recruit Employee/Manager** -- Recruit and activate new employees or managers.
  2. **Terminate Employee** -- Search and terminate an employee/user account.
  3. **Promote Employee** -- Promote an employee to an upper position.
  4. **Update Employee Information** -- Search and update employee details.
  5. **Finalize Leave Requests** -- Approve or reject leave requests forwarded by managers/employees.

N.B: Each user type must have at least 3 distinct features different from the others. Additional features may be added if desired. Proposals with generic, incomplete, or duplicated features across user types will receive reduced marks under the criteria below.